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PRODUCT GUIDELINE

LABORATORY FILTRATION

Focus on High-Precision Filtration



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COMPANY PROFILE

Since 2015, Orientfiltr Technology Co., Ltd has been specializing in advanced filtration solutions, providing high-quality filters, membranes, and housings to a diverse range of industries, including water treatment, food and beverage, oil and chemicals, electronics, petrochemical, seawater desalination, and power plants.

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Certification













Focus on High-Precision Filtration

Orientfiltr Technology Co., Ltd is a professional one-stop manufacturer and supplier of filtration solutions. We are dedicated to delivering high-quality filtration products that meet the diverse needs of customers worldwide. Our commitment extends to providing exceptional service while promoting health, safety, and environmentally sustainable development.

Syringe Filter

Application

1. Sample preparation for HPLC and UHPLC analysis.
2. Protect columns and extend column lifetimes.
3. Ensure compatibility with low sample volumes.
4. Maintain high integrity during sample preparation.
5. Color coding for easy identification and selection of the correct membrane and pore size.
6. Utilize integrated prefilter options for enhanced particulate removal.

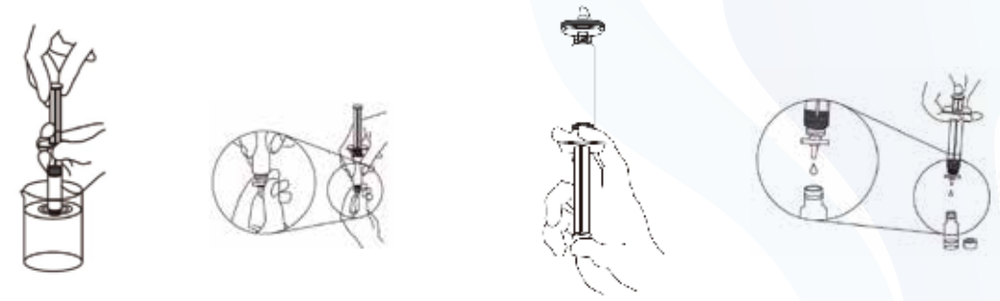
Membrane Compatibility

Sample Type	Membrane
Standard HPLC/samples/solvents	Nylon (NY)/ RC/PTFE
Standard GC samples/solvents	PTFE
UV spectrometry	PVDF
Capillary electrophoresis	PES
High particle load	Glass Fiber (GF)
Protein analysis	PVDF
Protein removal	GF
Trace metals	PES
Aggressive and nonpolar solvent	PTFE
Biological sample preparation	CA
Protein and peptide samples, general HPLC	Polypropylene (PP)
Highly particulates samples	GF

Sample Volume

Sample Volume	Diameter
≤120mL	30mm
≤100mL	25mm
≤20mL	17mm
≤10mL	13mm
≤1mL	4mm

How to Use Syringe Filters



PVDF (Hydrophilic/Hydrophobic) Syringe Filters

Superior membrane quality for HPLC. Hydrophilic PVDF syringe filters do not require pre-wetting for use with aqueous samples. Hydrophobic PVDF filters require pre-wetting with an alcohol before use with aqueous samples.

- Low non-specific binding with exceptional chemical resistance.
- Hydrophilic membrane provides excellent flow rates and low binding coefficients.
- Certified for chromatographic performance.
- Compatible with a wide range of aqueous and organic-based sample environments.
- Recommended for HPLC and UPLC. PVDF syringe filters have a well defined pore structure, giving effective retention of particles without excessive pore blockage. Compatible with aqueous and most organic solvents.



Application

1. Preparation and cleanup of samples for HPLC and organic solvents.
2. Handling protein-based samples with high non-specific binding.
3. Processing environmental water samples.

Specifications of PVDF Syringe Filters

Membrane	HPLC certified PVDF(Hydrophilic/Hydrophobic)
Max. Operating Temperature	<100 °C
Housing	Medical grade, virgin polypropylene
Pore Size	0.22 µm, 0.45 µm
Chemical Incompatibilities	DMF, DMSO, MEK, acetone and most caustic solutions > 6N

Regenerated Cellulose (RC) Syringe Filters

Superior chemical resistance, optimized for biological sample recoveries.

- Hydrophilic membrane provides excellent flow rates and extremely low binding coefficients
- Superior choice for biological assays, gel capsule dissolution testing, protein sample matrixes
- Compatible with a wide range of aqueous and organic-based sample environments
- Recommended for reverse phase and normal phase HPLC. Regenerated cellulose syringe filters are ideally suited for almost any laboratory procedure, from HPLC sample preparation to dissolution sample testing. Regenerated cellulose possesses superior chemical resistance.

Application

1. Preparation and cleanup of HPLC and organic solvent samples.
2. Analysis of dissolution samples, particularly for high-binding tablets or capsules.
3. Handling protein-based samples.
4. Sample analysis requiring maximum recovery.
5. Analyses that require low non-specific binding across a wide pH range.

Specifications of RC Syringe Filters

Membrane	Regenerated Cellulose
Max. Operating Temperature	<100 °C
Housing	Medical grade, virgin polypropylene
Pore Size	0.22 µm, 0.45 µm
Chemical Incompatibilities	Acetic acid, Acetone, Amyl acetate, Cyclohexanone, Sulfuric acid, Sodium hydroxide

Cellulose Acetate Syringe Filters

For filtering of aqueous solutions or biological samples.

- Hydrophilic membrane provides excellent flow rates and extremely low protein-binding coefficients.
- Superior choice for biological assays, gel capsule dissolution testing, protein sample matrixes.
- Recommended for aqueous last point change to —> Recommended for aqueous HPLC applications, this physically strong membrane is suitable for use with heated liquids. physically strong membrane which can be used with heated liquids.

Application

1. Protein-based samples with high non-specific binding.
2. Sample analysis which require maximum recoveries.

Specifications of CA Syringe Filters

Membrane	Cellulose Acetate
Max. Operating Temperature	<60 °C
Housing	Medical grade, virgin polypropylene
Pore Size	0.22 μm, 0.45 μm
Chemical Incompatibilities	Acetic acid, Acetone, Amyl acetate, Cyclohexanone, Sulfuric acid, Sodium ,hydroxide, 6N

Glass Fiber (GF) Syringe Filters

- For large particulate removal.
- Increased sample throughput.
- Low extraction neutral borosilicate glass.

Application

1. Clarification.
2. Pre-filtering of suspensions.

Specifications of Glass Fiber Syringe Filters

Membrane	Binder free Glass fiber
Max. Operating Temperature	<150°C
Housing	Medical grade, virgin polypropylene
Pore Size	0.7 μm, 1.0 μm
Chemical Incompatibilities	Hydrofluoric Acid,Sodium Hydroxide,Potassium Hydroxide

PTFE (Hydrophobic and Hydrophilic) Syringe Filters

Hydrophilic PTFE can be used for the same application but no pre-wetting of the membrane is required . PTFE syringe filters are applicable for filtration of gaseous or organic solvent-based samples. Both membrane types exhibits broad chemical resistance and unsurpassed temperature stability to address aggressive sample matrixes and extreme temperature situations. The hydrophobic PTFE filter can be utilized as a moisture barrier in venting applications. PTFE hydrophobic membranes require pre-treatment with alcohol before being suitable for aqueous or high aqueous/organic samples. Do not use directly with aqueous solutions.

- Excellent chemical resistance for use with organic matrices.
- Excellent flow rates and high loading capacities.
- Exceptional temperature stability.
- Organic solvent recommended.

Application

1. HPLC and organic solvent sample preparation and clean up.
2. Dissolution sample analysis.
3. General sample preparation prior to analysis
4. Elevated temperature samples, caustic or acidic solutions.

Specifications of PTFE Syringe Filters

Membrane	PTFE (Hydrophobic and Hydrophilic)
Max. Operating Temperature	<120°C
Housing	Medical grade, virgin polypropylene
Pore Size	0.22 μm, 0.45 μm
Chemical Incompatibilities	N/A

Nylon Syringe Filters

Hydrophilic nylon is extremely well suited for aqueous or organic sample preparation and HPLC, GC or dissolution sample analysis. Due to its excellent flow characteristics and mechanical stability, nylon offers the best combination of physical parameters to meet the most stringent analytical needs.

- Naturally hydrophilic with broad chemical resistance.
- Excellent flow rates and high-throughput loading.
- HPLC recommended.

Application

1. HPLC and organic solvent sample preparation and clean up.
2. Dissolution sample analysis.
3. General sample preparation prior to analysis.
4. Mixed sample matrix of aqueous or organic dissolved analytes.

Specifications of Nylon Syringe Filters

Membrane	HPLC certified nylon
Max. Operating Temperature	<100 °C
Housing	Medical grade, virgin polypropylene
Pore Size	0.22 μm, 0.45 μm
Chemical Incompatibilities	Formic Acid,Chloroform,Chlorobenzene,Cresol Cyclohexane, Dichloromethane

PES syringe filters — Ion Chromatography (IC) Certified

This hydrophilic polymer has excellent cleanliness and is compatible with a wide range of solvents. It is the membrane of choice for ion chromatography applications.

- Certified for low-level IC interference by ICP analysis.
- Hydrophilic membrane provides excellent flow rates and low binding coefficients.
- Low affinity for binding drugs, ideal for dissolution testing procedures.

Application

1. IC sample preparation and analysis.
2. Dissolution testing.

Specifications of IC PES Syringe Filters

Membrane	IC Certified PES
Max. Operating Temperature	<100 °C
Housing	Medical grade, virgin polypropylene
Pore Size	0.22 μm, 0.45 μm
Chemical Incompatibilities	Acetone,Acetonitrile,Chloroform,DMSO,MEK,Dichloromethane,Pyridine

Sterile Syringe Filter

Orientfiltr sterile syringe filters are made with a variety of membrane filters housed in polypropylene (PP) casings, utilizing the most advanced methods and design features. The most commonly used membranes for sterile syringe filters are PES and PES with a GF pre-filter. PES membranes offer high flow rates and low protein adsorption, making them suitable for a wide range of applications, such as buffers, proteins, culture media, additives, and more. The GF pre-filter provides excellent particle-holding capacity, significantly increasing the throughput for challenging samples.



Features

- Reliable sterile filtration, LRV>7 for Brevundimonas diminuta, ATCC19146.
- No pyrogen.
- Larger effective filtration area (EFA) of 4.9 cm^2.
- The lowest protein adsorption.
- Excellent flow rate.
- High total throughput.
- Hold-up volume less than 120μL.
- Individually packaged and sterilized by Gamma Radiation.

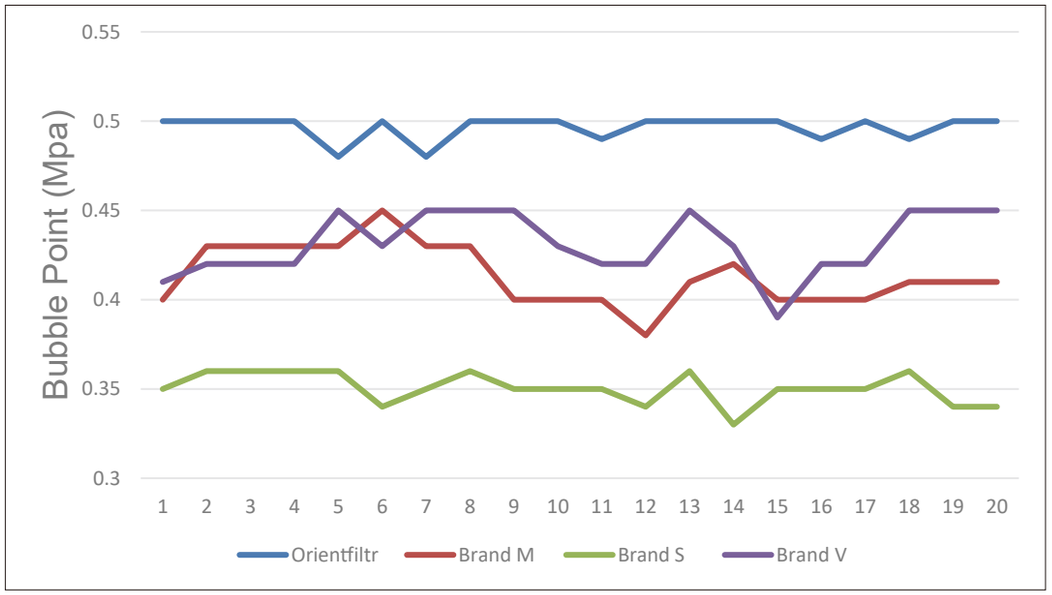
Application

1. Ideal for rapid filtration of serum-based tissue culture media, biological fluids, fixation buffers, and other challenging solutions.
2. Fast sterilization of buffers, culture media or other solutions (PES 0.22 only).
3. Clarification and prefiltration of aqueous solutions (PES 0.45).

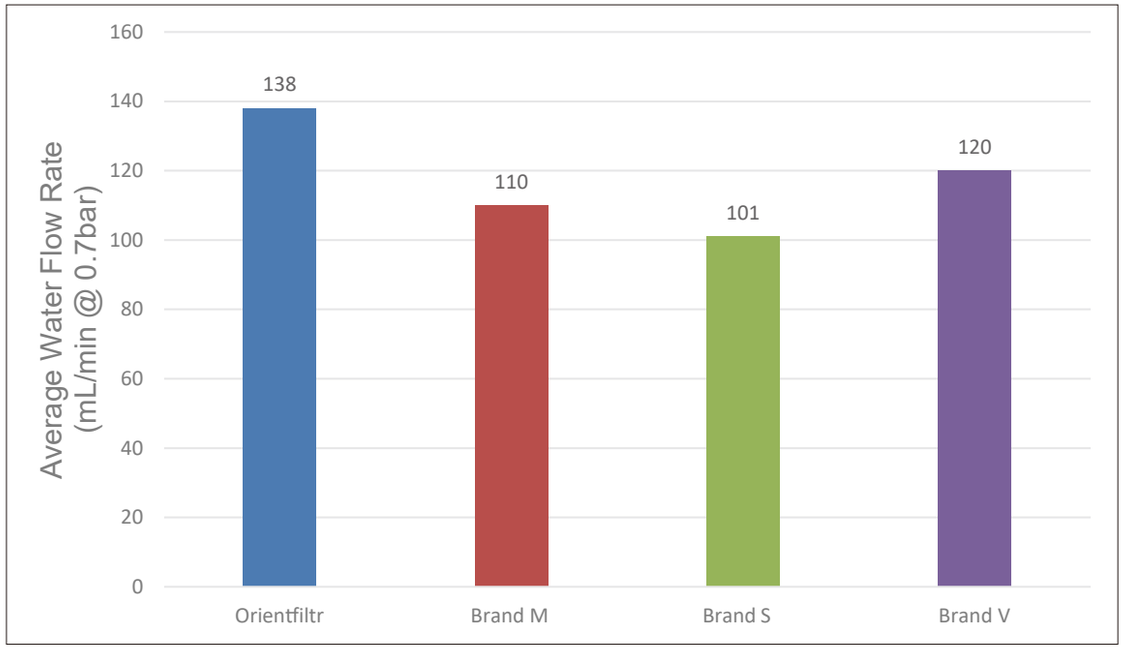
Cross Reference

Description	Brand M	Brand V	Brand S	Orientfiltr PN
Sterile PES Syringe Filter,0.2μm,33mm	SLGP033RB (250pcs)	76479 - 016(30mm)	16532————GUK	IGPES3322S
	SLGP033RK (1000pcs)		16532————K	
	SLGP033RS (50pcs)		16541————K	
Sterile PES Syringe Filter,0.45μm,33mm Sterile PVDF	SLHP033RB (250pcs)	76479 - 012(30mm)	16537————K	IGPES3345S
	SLHP033RS (50pcs)		16533————GUK	
			16533————K	
Syringe Filter,0.2μm,33mm	SLGV033RS (50pcs)	76478 - 988(30mm)		IGPVL3322S
	SLGV033RK (1000pcs)			

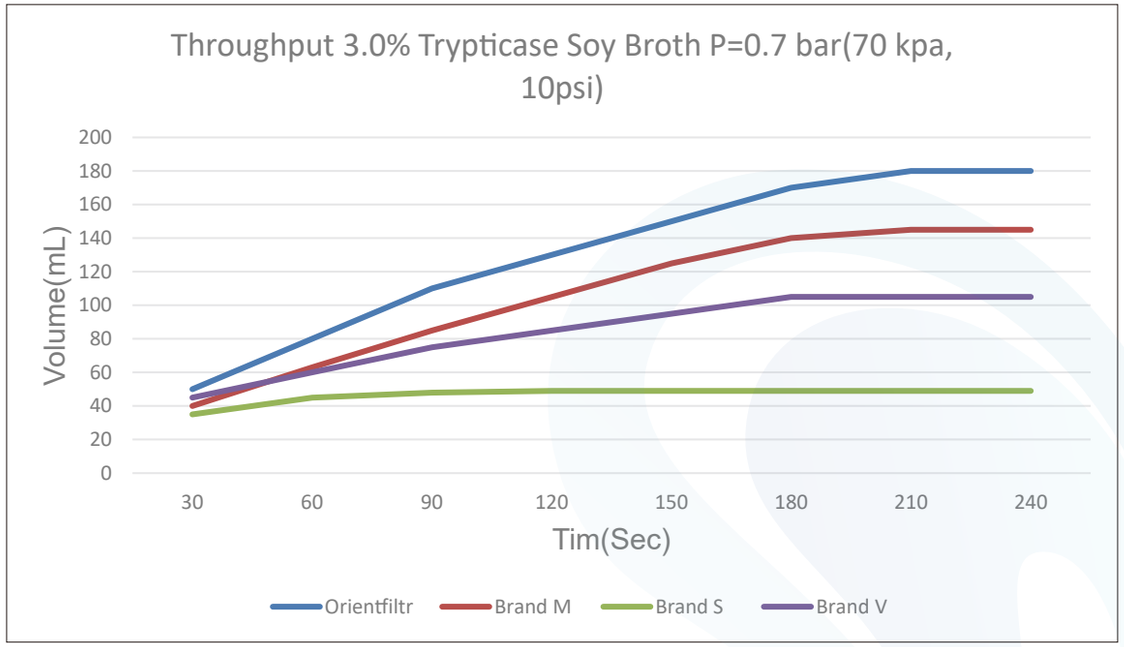
High and Stable Bubble Point



High Water Flow Rate



High Throughput



Ordering Information

Part Number	Description	Packing
IGPES3322S	Sterile Syringe Filter, PES, 33mm, 0.22μm	100pcs
IGPES3345S	Sterile Syringe Filter, PES, 33mm, 0.45μm	100pcs
IGPVL3322S	Sterile Syringe Filter, PVDF, 33mm, 0.22μm	100pcs
IGPVL3345S	Sterile Syringe Filter, PVDF, 33mm, 0.45μm	100pcs
IGCA3322S	Sterile Syringe Filter, CA, 33mm, 0.22μm	100pcs
IGCA3345S	Sterile Syringe Filter, CA, 33mm, 0.45μm	100pcs
WPES3322S	Welding Sterile Syringe Filter, PES, 33mm, 0.22μm	100pcs
WPES3345S	Welding Sterile Syringe Filter, PES, 33mm, 0.45μm	100pcs
WPVL3322S	Welding Sterile Syringe Filter, PVDF, 33mm, 0.22μm	100pcs
WPVL3345S	Welding Sterile Syringe Filter, PVDF, 33mm, 0.45μm	100pcs
WCA3322S	Welding Sterile Syringe Filter, CA, 33mm, 0.22μm	100pcs
WCA3345S	Welding Sterile Syringe Filter,CA, 33mm, 0.45μm	100pcs

50mm Disc Capsule Filter

Orientfiltr Sterile thermally bonded 50mm venting filter incorporates a hydrophobic PTFE membrane in a polypropylene housing. It is designed for sterile venting, gas, and liquid filtration.
Its material of construction and design provide excellent thermal and chemical compatibility. Orientfiltr venting filter is available in 0.2 μm and 0.45 μm pore sizes.



Features

- 100% Integrity test.
- Robust design and construction allow for multiple autoclave cycles>20.
- Designed to be reusable but also offers the convenience of being disposable.
- Available in pre-sterilized or non-sterilized packaging.
- PTFE membrane and polypropylene housing offers broad chemical compatibility.
- Optimized PTFE membrane provides superior flow rates in a compact device.
- Provides ULPA filtration (>99.999% @ 0.1um).
- Lightweight design prevents tube pinching in carboy venting applications.
- For liquid sterile filtration, we also offer support with PES membranes and hydrophilic PVDF membranes.

Application

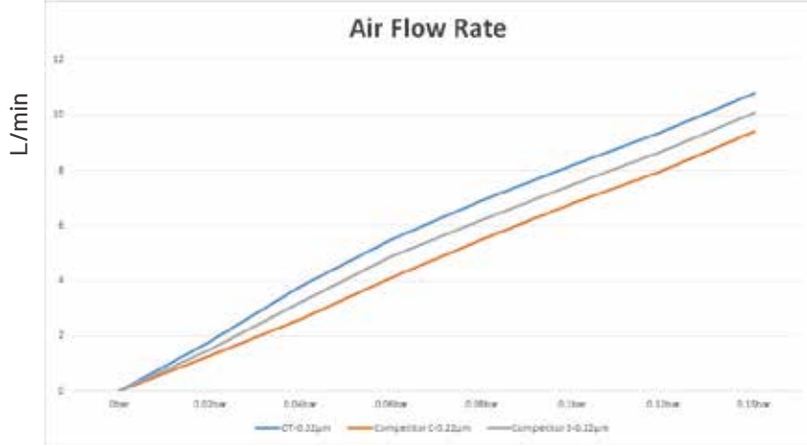
1. Sterile Venting of bioreactors, fermentation tanks, media flasks, and carboys.
2. Sterile gas purge of cell culture vessels or filling vessels.
3. In-line sterilization of and particulate removal from air and gases.

Specifications

Housing Material	Polypropylene
Filter Material	Hydrphobic PTFE/PES/Hydrophilic PVDF

Sealing Technology	Thermally bonded
Housing Diameter	64 mm
Filtration Area	20 cm2
Inlet/Outlet Connections	1/4-3/8" (7-13mm) stepped HB
Pore Size (liquid rating)	0.2 μm/ 0.45 μm
Bubble Point with Isopropanol (60%)	0.2 μm, > 14.5psi (1.0bar)/ 0.45 μm, > 7.25psi (0.5bar)
Maximum Operating Pressure	43.5psi (3.0bar)
Housing Burst Pressure	72.5 psi (5.0 bar)
Max. Autoclaving Temperature	121℃ (249.8 ℉)
Autoclave Cycles	≥20
Hold-up Volume	Before bubble point approx. 1.0 mL After bubble point approx. 0.5 mL
Sterilization	EO sterilization/Non-sterile
PTFE Air Flow Rate @ p=2.9psi	0.2μm: 5 lpm; 0.45 μm: 8 lpm

Air Flow Rate



Ordering Information

Part Number	Description	Packing
VPTB5022N	Non-Sterile Venting Filter, PTFE, 50mm, 0.22μm	20pcs
VPTB5045N	Non-Sterile Venting Filter, PTFE, 50mm, 0.45μm	20pcs
VPTB5022S	Sterile Venting Filter, PTFE, 50mm, 0.22μm	20pcs
VPTB5045S	Sterile Venting Filter, PTFE, 50mm, 0.45μm	20pcs

Gridded Membrane Filter

Orientfiltr mixed cellulose ester (MCE) and cellulose nitrate membrane filters are ideal for colony counting and sterility testing. Orientfiltr MCE membranes are composed of cellulose acetate and cellulose nitrate, making them one of the most widely used membranes in analytical and research applications. Our high-performance MCE membranes use microbiologically "friendly" ink on grids, promoting growth to maximize recovery.



Features

- High recovery rates for microorganisms (>90%).
- Bacterial Challenge: LRV>7 (0.2um, ATCC 19146, ASTM bacterial challenge test) for each lot of products.
- High flow rate membrane.
- Easy counting of colonies.
- Ready-to-use packages.
- 5-step traceable QC inspections of manufacturing from raw material to finished products.
- Technical Support.
- Competitive pricing.

Application

1. Sterilizing filtration, Air monitoring, Particle monitoring.
2. Clarification of aqueous solution.
3. Particle removal and analysis.
4. Microbiology analysis.

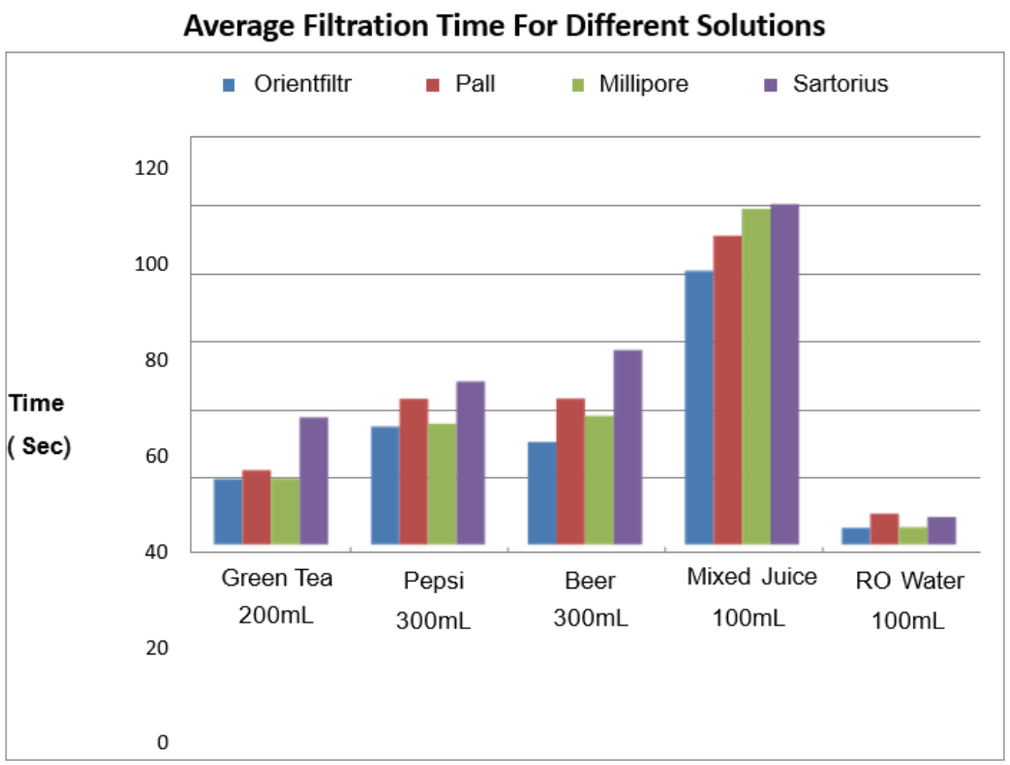
Wettability

	Orientfiltr	Pall	Millipore	Sartorius	Advantec
Wettability (Sec)	1-2	60-120	5-6	8-10	2-4

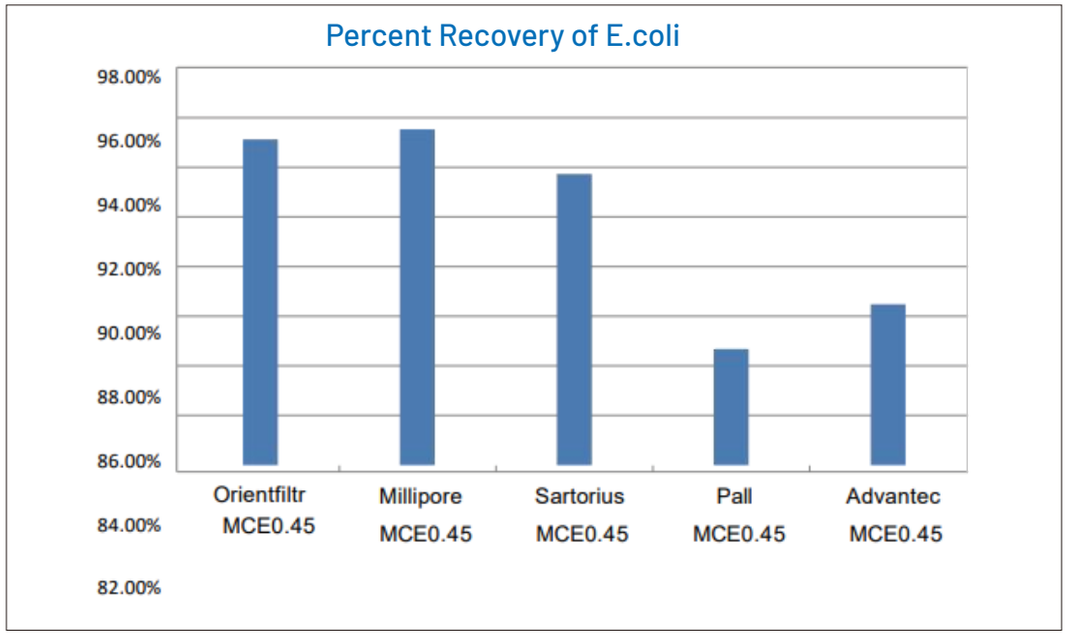
Bacterial Challenge

	Orientfiltr	Pall	Millipore	Sartorius	Advantec
Bacterial challenge	Pass	Pass	Pass	Pass	Pass

	Orientfiltr	Pall	Millipore	Sartorius	Advantec
Part Number	GMCE4745WS	63020	HAWG04700	11406—47—ACN	A045F047A



Rate of E.coli Recovery



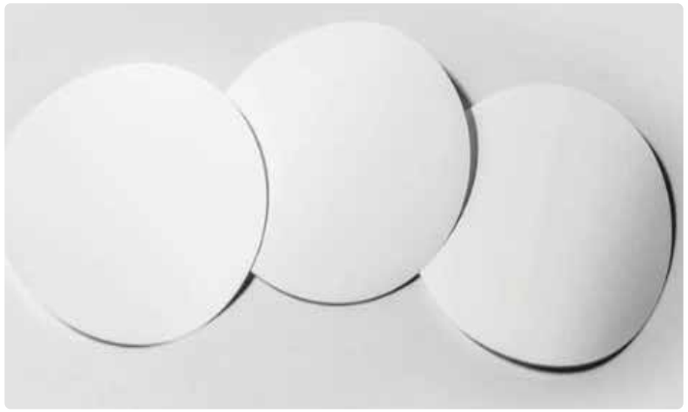
Order Information

Part Number	Description	Packing
GMCE4722WS	White, Individually package, Sterile, 0.22μm, 47mm	100pcs
GMCE4745WS	White, Individually package, Sterile, 0.45μm, 47mm	100pcs
GMCE5022WS	White, Individually package, Sterile, 0.22μm, 50mm	100pcs
GMCE5045WS	White, Individually package, Sterile, 0.45μm, 50mm	100pcs

Glass Fiber Membrane

Environmental Analysis Glass Fiber Membrane

Monitoring contaminants in air, water, and soil, while gaining deeper insights into their effects on human and animal health, is both vital and frequently difficult. As environmental pollutants face tighter regulations, the need for reliable methods that consistently deliver high-quality data, like those utilized by the U.S. Environmental Protection Agency, continues to grow.
Orientfiltr Glass Fiber membrane is widely used in environmental analysis, particularly for sampling and filtering contaminants in air, water, and other environmental matrices. Its unique properties make it highly suitable for capturing particles and pollutants in various applications.



Features

- High flow rate.
- High dirt-holding capacity.
- Excellent chemical compatibility.
- Binder-free.

Application

1. Air Quality Monitoring
2. Chemical and Analytical Testing
3. Soil and Sediment Analysis
4. Trace Metal Analysis

Specifications

Part Number	Pore Size (μm)	Weight (g/m²)	Thickness(μm)
CGF47160	1.6	53	250-310
CGF47100	1	143	710-830
CGF47120	1.2	53	260-320
CGF47270	2.7	121	580-640
CGF4770	0.7	75	450-520
CGF47150	1.5	70	350-440

Order Information

Part Number	Description
CGF47160	Glass Fiber membrane disc; 47mm diameter; 1.6μm
CGF47100	Glass Fiber membrane disc; 47mm diameter; 1.0μm
CGF47120	Glass Fiber membrane disc; 47mm diameter; 1.2μm
CGF47270	Glass Fiber membrane disc; 47mm diameter; 2.7μm
CGF4770	Glass Fiber membrane disc; 47mm diameter; 0.7μm
CGF47150	Glass Fiber membrane disc; 47mm diameter; 1.5μm

Other available diameters: 13/25/47/90/142mm
Customized formats available

Disc Membrane Filter

Filtration membrane discs are thin, precut membrane rounds designed for high-throughput, and fast filtration of liquid and gas samples in analytical research. This is one kind of physical filtration process where a contaminated fluid is passed through a special pore size membrane to separate micro-organisms and suspended particles from process liquid or gas. The typical pore size used for micro-filtration ranges from about 0.1 to 10μm.
Orientfiltr offers various disc membrane filters for liquid filtration and air filtration, including, Nylon, PES, PVDF, PTFE, PP, MCE, CN, and CA. The disc diameters are from 13mm to 293mm. We can also customize the diameter for you.

Features

- Low HPLC extractables.
- High flow rate and high filtration efficiency.
- Lot TOC level (environmental PES).
- High air flow rate (Hydrophobic PTFE).
- Sterile(Gamma/EO) available.

Application

1. General clean and filtration
2. Sample Preparation
3. Filtration of Aqueous and organic solutions
4. Chromatography and Mass Spec Sample Preparation
5. Environmental Monitoring and Analysis
6. Venting and Gas Filtration

Specifications

Membrane Type	Pore Size(um)	Diameter(mm)
PES	0.1-3.0	13, 25, 37, 47, 90, 142, 293
MCE	0.1-8.0	13, 25, 37, 47, 90, 142
NYLON	0.1- 10.0	13, 25, 37, 47, 90, 142, 293
PVDF(Hydrophobic/Hydrophilic)	0.1-0.45	13, 25, 37, 47, 90, 142
PTFE(Hydrophobic/Hydrophilic)	0.1-5.0	13, 25, 37, 47, 90, 142
PP	0.22-20.0	13, 37, 47, 90, 125, 142, 293
CA	0.22-8.0	13, 25, 37, 47, 90, 142

